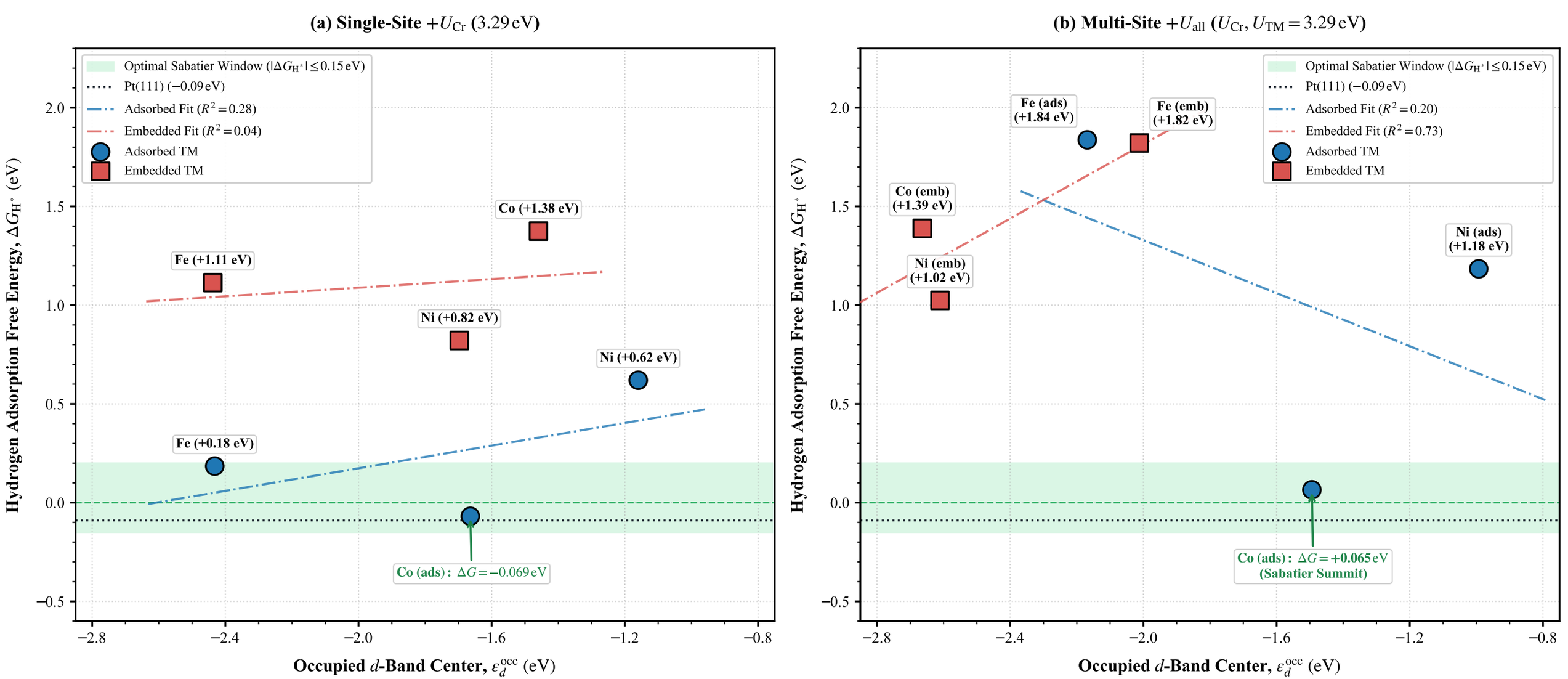

(a) Single-Site +U_{Cr} (3.29 eV)
(b) Multi-Site +U_{all} (U_{Cr}, U_{TM} = 3.29 eV)
Optimal Sabatier Window (|ΔG_{H^*}| ≤ 0.15 eV)
Pt(111) (−0.09 eV)
Adsorbed Fit (R^2 = 0.28)
Embedded Fit (R^2 = 0.04)
Adsorbed TM
Embedded TM
Fe (+1.11 eV)
Co (+1.38 eV)
Ni (+0.82 eV)
Ni (+0.62 eV)
Fe (+0.18 eV)
Co (ads): ΔG = −0.069 eV
Adsorbed Fit (R^2 = 0.20)
Embedded Fit (R^2 = 0.73)
Fe (ads) (+1.84 eV)
Fe (emb) (+1.82 eV)
Co (emb) (+1.39 eV)
Ni (emb) (+1.02 eV)
Ni (ads) (+1.18 eV)
Co (ads): ΔG = +0.065 eV (Sabatier Summit)
Hydrogen Adsorption Free Energy, ΔG_{H^*} (eV)
Occupied d-Band Center, ε_d^{occ} (eV)